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Assignment 1

Computer Networks

I-409-TSAM

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Task 1

A

The open TCP ports were:
22/tcp – SSH
2000/tcp – Cisco SCCP
5060/tcp – SIP

```
evie@evies-MacBook-Air ~ % sudo nmap -sS 130.208.246.98
Password:
Starting Nmap 7.991 ( https://nmap.org ) at 2026-08-30 15:23 +0000
Nmap scan report for 130.208.246.98
Host is up (0.0068s latency).
Not shown: 880 filtered tcp ports (no-response), 117 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
2000/tcp open  cisco-sccp
5060/tcp open  sip

Nmap done: 1 IP address (1 host up) scanned in 4.21 seconds
evie@evies-MacBook-Air ~ %
```

ping 130.208.246.98 works and responds with 64 byte packages taking 2-10milliseconds each depending on computer used.

```
evie@evies-MacBook-Air ~ % ping 130.208.246.98
PING 130.208.246.98 (130.208.246.98): 56 data bytes
64 bytes from 130.208.246.98: icmp_seq=0 ttl=50 time=8.936 ms
64 bytes from 130.208.246.98: icmp_seq=1 ttl=50 time=9.273 ms
64 bytes from 130.208.246.98: icmp_seq=2 ttl=50 time=9.549 ms
64 bytes from 130.208.246.98: icmp_seq=3 ttl=50 time=9.560 ms
64 bytes from 130.208.246.98: icmp_seq=4 ttl=50 time=4.622 ms
64 bytes from 130.208.246.98: icmp_seq=5 ttl=50 time=4.586 ms
64 bytes from 130.208.246.98: icmp_seq=6 ttl=50 time=5.490 ms
64 bytes from 130.208.246.98: icmp_seq=7 ttl=50 time=9.274 ms
^C
--- 130.208.246.98 ping statistics ---
8 packets transmitted, 8 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 4.586/7.661/9.560/2.162 ms
evie@evies-MacBook-Air ~ %
```

B

```
tcpdump: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
Remember to check for network connectivity
15:43:11.118241 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 64)
    192.168.1.137.49466 > 130.208.246.98.commplex-main: Flags [SEW], cksum 0x1d84 (correct), seq 560349985, win 65535, options [mss 1460,nop,wscale 6,nop,nop,TS val 867987533 ecr 0,sackOK,eol], length 0
        0x0000:  4500 0040 0000 4000 4006 ff53 c0a8 0189  E..@..@.@..S....
        0x0010:  82d0 f662 c13a 1388 2166 4321 0000 0000  ...b.:..!fC!....
        0x0020:  b0c2 ffff 1d84 0000 0204 05b4 0103 0306  ................
        0x0030:  0101 080a 33bc 704d 0000 0000 0402 0000  ....3.pM........
15:43:11.122936 IP (tos 0x0, ttl 50, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    130.208.246.98.commplex-main > 192.168.1.137.49466: Flags [S.E], cksum 0xbfdd (correct), seq 143520767, ack 560349986, win 65160, options [mss 1412,sackOK,TS val 1722224012 ecr 867987533,nop,wscale 7], length 0
        0x0000:  4500 003c 0000 4000 3206 0d58 82d0 f662  E..<..@.2..X...b
        0x0010:  c0a8 0189 1388 c13a 088d f3ff 2166 4322  .......:....!fC"
        0x0020:  a052 fe88 bfdd 0000 0204 0584 0402 080a  .R..............
        0x0030:  66a7 0d8c 33bc 704d 0103 0307            f...3.pM....
15:43:11.123039 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 52)
    192.168.1.137.49466 > 130.208.246.98.commplex-main: Flags [.], cksum 0xe534 (correct), ack 1, win 2057, options [nop,nop,TS val 867987538 ecr 1722224012], length 0
        0x0000:  4500 0034 0000 4000 4006 ff5f c0a8 0189  E..4..@.@.._....
        0x0010:  82d0 f662 c13a 1388 2166 4322 088d f400  ...b.:..!fC"....
        0x0020:  8010 0809 e534 0000 0101 080a 33bc 7052  .....4......3.pR
        0x0030:  66a7 0d8c                                f...
15:43:11.128830 IP (tos 0x2,ECT(0), ttl 50, id 4050, offset 0, flags [DF], proto TCP (6), length 53)
    130.208.246.98.commplex-main > 192.168.1.137.49466: Flags [P.], cksum 0xe131 (correct), seq 1:2, ack 1, win 510, options [nop,nop,TS val 1722224017 ecr 867987538], length 1
        0x0000:  4502 0035 0fd2 4000 3206 fd8a 82d0 f662  E..5..@.2......b
        0x0010:  c0a8 0189 1388 c13a 088d f400 2166 4322  .......:....!fC"
        0x0020:  8018 01fe e131 0000 0101 080a 66a7 0d91  .....1......f...
        0x0030:  33bc 7052 0a                             3.pR.
15:43:11.128939 IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 52)
    192.168.1.137.49466 > 130.208.246.98.commplex-main: Flags [.], cksum 0xe528 (correct), ack 2, win 2057, options [nop,nop,TS val 867987544 ecr 1722224017], length 0
        0x0000:  4500 0034 0000 4000 4006 ff5f c0a8 0189  E..4..@.@.._....
        0x0010:  82d0 f662 c13a 1388 2166 4322 088d f401  ...b.:..!fC"....
        0x0020:  8010 0809 e528 0000 0101 080a 33bc 7058  .....(......3.pX
        0x0030:  66a7 0d91                                f...
15:43:23.413093 IP (tos 0x2,ECT(0), ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 95)
    192.168.1.137.49466 > 130.208.246.98.commplex-main: Flags [P.], cksum 0x284c (correct), seq 1:44, ack 2, win 2057, options [nop,nop,TS val 867999828 ecr 1722224017], length 43
        0x0000:  4502 005f 0000 4000 4006 ff32 c0a8 0189  E.._..@.@..2....
        0x0010:  82d0 f662 c13a 1388 2166 4322 088d f401  ...b.:..!fC"....
        0x0020:  8018 0809 284c 0000 0101 080a 33bc a054  ....(L......3..T
        0x0030:  66a7 0d91 5265 6d65 6d62 6572 2074 6f20  f...Remember.to.
        0x0040:  6368 6563 6b20 666f 7220 6e65 7477 6f72  check.for.networ
        0x0050:  6b20 636f 6e6e 6563 7469 7669 7479 0a    k.connectivity.
15:43:23.417154 IP (tos 0x2,ECT(0), ttl 50, id 4051, offset 0, flags [DF], proto TCP (6), length 52)
    130.208.246.98.commplex-main > 192.168.1.137.49466: Flags [.], cksum 0x8b0b (correct), ack 44, win 510, options [nop,nop,TS val 1722236306 ecr 867999828], length 0
        0x0000:  4502 0034 0fd3 4000 3206 fd8a 82d0 f662  E..4..@.2......b
        0x0010:  c0a8 0189 1388 c13a 088d f401 2166 434d  .......:....!fCM
        0x0020:  8010 01fe 8b0b 0000 0101 080a 66a7 3d92  ............f.=.
        0x0030:  33bc a054                                3..T
```

C

sudo tcpdump −vAX −i ens192 ’port 5000’ captures or monitors network traffic or packets passing through a network interface of our choice. v is verbose, A and X are both in what kind of print format we want it, -i interface is what network interface it is, port is what port we are using

ncat −ln 5000 starts a listening server that waits for a "connection", -l is listen, -n is is using a raw IP address to make it faster. 5000 is the port

ncat 130.208.246.98 5000 this is the client, the ip address is the destination, and the port is the port. the command estabilishes an outbound connection to server on port 5000.

D

```
evie@evies-MacBook-Air ~ % sudo tcpdump -vAX -i en0 'udp and port 5000'
Password:
tcpdump: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
15:49:06.121622 IP (tos 0x0, ttl 64, id 52977, offset 0, flags [none], proto UDP (17), length 71)
    192.168.1.137.50237 > 130.208.246.98.commplex-main: UDP, length 43
        0x0000:  4500 0047 cef1 0000 4011 7050 c0a8 0189  E..G....@.pP....
        0x0010:  82d0 f662 c43d 1388 0033 5fb0 5265 6d65  ...b.=...3_.Reme
        0x0020:  6d62 6572 2074 6f20 6368 6563 6b20 666f  mber.to.check.fo
        0x0030:  7220 6e65 7477 6f72 6b20 636f 6e6e 6563  r.network.connec
        0x0040:  7469 7669 7479 0a                        tivity.
```

E

The TCP capture contained several packets because TCP first establishes a connection using a three-way handshake (SYN, SYN-ACK, ACK), and then uses acknowledgements when data is sent. In the UDP capture, there was no handshake; the message was sent directly in a UDP datagram. This difference occurs because TCP is connection-oriented, while UDP is connectionless.

Task 2

Code in scanner.c

The scanner found these open UDP ports:

- 4013
- 4044
- 4081
- 4052 THE EVIL PORT

We compared the result with Nmap using sudo nmap -sU -p 4000-4100 130.208.246.98. Nmap also reported ports 4013, 4044, and 4081 as open. Nmap additionally reported port 4052 as open|filtered, meaning it could not determine whether the port was open or filtered. Further testing showed it open as well. Example of scan

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-31 14:59 GMT
Warning: 130.208.246.98 giving up on port because retransmission cap hit (3).
Nmap scan report for 130.208.246.98
Host is up (0.0061s latency).
Not shown: 82 open|filtered udp ports (no-response)
PORT     STATE  SERVICE
4012/udp closed pda-gate
4013/udp open   acl-manager
4016/udp closed talarian-mcast2
4020/udp closed trap
4025/udp closed partimage
4028/udp closed dtserver-port
4030/udp closed jdmn-port
4035/udp closed wap-push-http
4038/udp closed fazzt-ptp
4044/udp open   ltp
4052/udp open   interact
4067/udp closed idp
4071/udp closed aibkup
4073/udp closed iRAPP
4075/udp closed perimlan
4077/udp closed ascomalarm
4081/udp open   lorica-in-sec
4083/udp closed lorica-out-sec
4097/udp closed patrolview

Nmap done: 1 IP address (1 host up) scanned in 8.02 seconds
```